Geospatial Foundation Models - Project Report

Finetuning Geospatial Foundation Models for Classification

by

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Repo Link (request access):
https://github.com/titocapovilla/Pia-GeoSpatial
Dataset Link: Google Drive Link

Geospatial Foundation Models - Polimi

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Abstract

This project explores the fine-tuning of geospatial foundation models for land cover classification using the LUCAS (Land Use and Coverage Area frame Survey) dataset, specifically focusing on a subset of satellite imagery chips from Italy. The primary objective was to evaluate the transfer learning capabilities of geospatial foundation models, when applied to a small, highly imbalanced dataset of $n \approx 2,000$ multispectral images across 10 distinct land cover categories. Terramind was chosen to complete the task.

The methodology involved a comprehensive data preparation pipeline, including the retrieval of Sentinel-2-derived imagery, statistical normalization, and a stratified train-validation split to mitigate class imbalance.

A deep learning architecture was constructed using the TerraMind encoder coupled with different decoders to process multispectral inputs. To address the scarcity of samples in minority classes (e.g., inland water and artificial surfaces), a class-weighted Cross-Entropy loss function and reduced weight-decay regularization were employed. The final four runs, discussed in this report compared Cross-Entropy loss with Focal loss.

After preliminary testing with smaller models to ensure stability, we focused on four final configurations. These experiments evaluated the impact of decoder complexity and compared Class-Weighted Cross-Entropy against Focal Loss.

Experimental results demonstrated a clear trade-off between loss minimization and minority class recall. While initial epochs showed higher raw accuracy across classes, the model ultimately converged toward a conservative, narrower strategy. This shift significantly reduced loss and improved Micro-Accuracy, leading to robust performance on dominant classes such as Arable Lands and Wooden Areas (achieving nearly 80% accuracy). However, this optimization exacerbated class imbalance issues, as the model learned to suppress predictions for spectrally similar minority vegetation subtypes rather than risk confident errors. This underscores the necessity of class balancing techniques when applying foundation models to highly skewed geospatial datasets, as well as the need for larger datasets that cover all classes with enough samples.

Keywords: Geospatial Foundation Models, TerraMind, Land Cover Classification, LUCAS Dataset, Transfer Learning, Sentinel-2, Multispectral Imagery, Class Imbalance, Fine-tuning, Remote Sensing, Deep Learning.

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1. Dataset Preparation

1 Size, Classes and representation

The provided dataset for this lab was a collection of images from the Lucas (Land Use and Coverage Area frame Survey) [1] dataset which was prepared for the Geospatial Foundation Models course at Polimi. The downloaded subset consists of 1,994 multispectral image chips stored as .tif files.

Key characteristics of the dataset include:

- **Source Data:** The images are derived from Sentinel-2 satellite data, containing 6 spectral bands: Blue, Green, Red, NIR Narrow, SWIR 1, and SWIR 2 (inferred from the normalization statistics and model configuration).
- **Classification:** The chips are labeled into 10 distinct land cover categories (based on the STR18 code), ranging from Arable Lands and Wooden Areas to Artificial constructions and Water bodies.
- **Distribution:** The dataset is notably imbalanced, resulting in our subset having classes like Wooden Areas (704 samples) and Arable Lands (511 samples) significantly outnumbering rare classes like Transitional waters (20 samples).

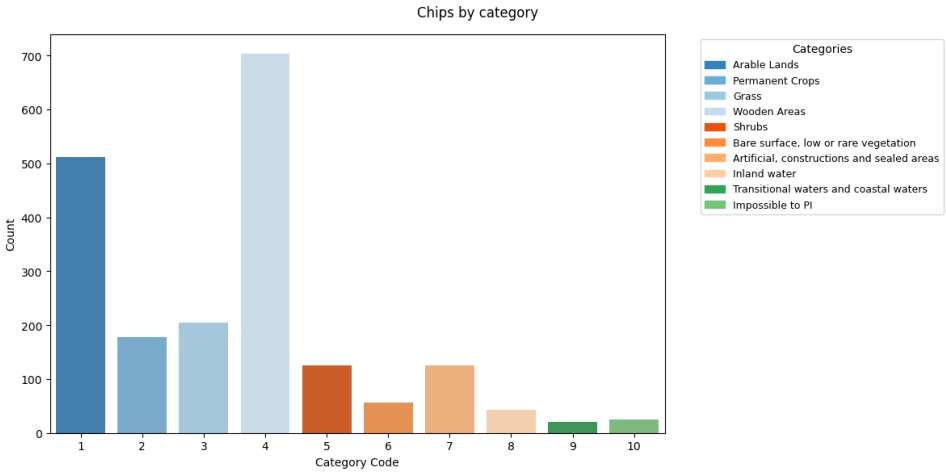


Figure 1.1: Dataset distribution

In order to preserve the dataset structure, we had to make decisions at the training stages to compensate for the highly imbalanced nature of the classes. Trimming all classes to the same size would have resulted in very small training sets per class, and a dataset that does not reflect the true data distribution and real-world land cover prevalence. Data was split into Trainining and Validation sets. The lack of a separate Test set was a mandatory choice due to the small sample size in some of the categories.

2. Training

1 Model and Size

TerraMind [2] was the chosen model to perform the land cover classification task. We selected this foundation model due to its ease of use and advanced geospatial capabilities, as well as its architectural potential to support multi-modal reasoning ('thinking') in future experimental phases. The training process followed a staged approach: TerraMind-v1-tiny was utilized for initial experimentation and rapid prototyping, while the TerraMind-v1-base variant was deployed for the final training runs to maximize performance

To determine the optimal configuration for land cover classification on the imbalanced LUCAS dataset, after several testing runs, we conducted a comparative analysis comprising four distinct experimental runs. These runs were designed to isolate two critical variables: the complexity of the decoder architecture and the efficacy of the loss function in handling class imbalance.

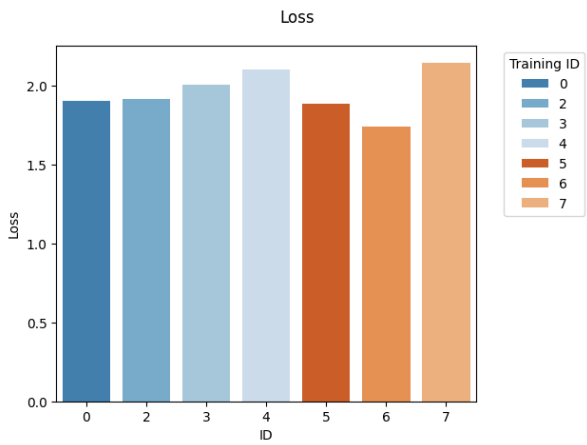


Figure 2.1: Early Experimental runs' suboptimal loss

1.1 Global Hyperparameters

A standardized set of hyperparameters was maintained across all experimental runs. All models were trained for 100 epochs. Optimization was performed using the AdamW optimizer with a fixed learning rate of 1e-4.

Regularization was adjusted by reducing the weight decay coefficient to 0.05. This lower penalty seemed to be effective in providing the model with the necessary capacity to capture fine-grained spectral distinctions inherent in the diverse land cover categories. Finally, a batch size of 8 was enforced to accommodate GPU memory constraints while maintaining gradient stability.

1.2 Loss Functions

We evaluated two distinct loss objectives to address the dataset's heavy imbalance and compare their impact on optimization stability. First, we employed Class-Weighted Cross-Entropy (CE). This objective function explicitly incorporates calculated class weights derived from the inverse frequency of training samples, thereby enforcing a heavier penalty for misclassifying rare categories. Second, we tested Class-Weighted Focal Loss. This configuration utilizes the same inverse-frequency class weights but augments them with a focusing mechanism that dynamically down-weights high-confidence predictions. This approach was selected to evaluate if combining static class balancing with dynamic sample difficulty weighting yields better convergence properties than static weighting alone.

1.3 Decoder Architecture

The TerraMind backbone produces high-dimensional token representations. We assessed two methods for projecting these tokens into final classification logits. The Identity Decoder utilizes a lightweight configuration consisting of a `SelectIndices` neck to extract features from the final backbone layer, followed by an `AggregateTokens` operation using mean pooling. This simple approach just passes the data through. Alternatively, the MLP Decoder introduces higher capacity by employing a Multi-Layer Perceptron with a hidden dimension of 512 channels. This gives the model more room to learn on top of what the backbone provided. This could help learn spacial textures and distinguish similar subclasses better.

1.4 Experimental Configuration

The experimental design was factorized into four distinct runs, summarizing the combinations of decoder complexity and loss objectives (Table 2.1).

Table 2.1: Summary of Experimental Runs

Run ID	Decoder Architecture	Loss Function
Run 1	Identity (Lightweight)	Weighted CE
Run 2	MLP (High Capacity)	Weighted CE
Run 3	Identity (Lightweight)	Focal Loss
Run 4	Identity (Soft Weights)	Focal Loss

3. Findings

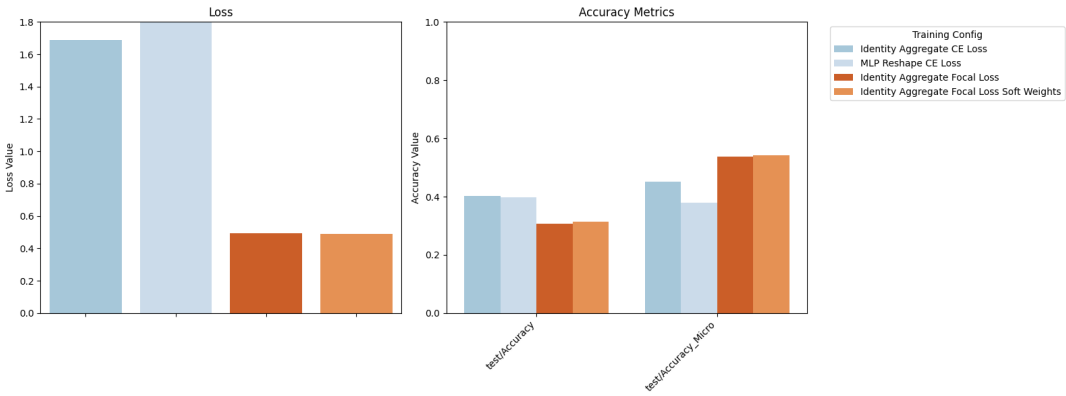
1 Findings and Results

The quantitative results from the four experimental runs are summarized in Table 3.1. The analysis reveals significant performance trade-offs driven primarily by the choice of loss function, while the decoder architecture played a negligible role.

Table 3.1: Quantitative Results on Test Set (100 Epochs)

Metric	**Run 1 (CE)**	**Run 2 (MLP+CE)**	**Run 3 (Focal)**	**Run 4 (Focal*)**
Test Loss	1.69	1.84	**0.49**	**0.49**
Macro Accuracy	**40.2%**	39.8%	30.6%	31.4%
Micro Accuracy	45.1%	37.8%	53.6%	**54.1%**

1.1 Loss Function Trade-off

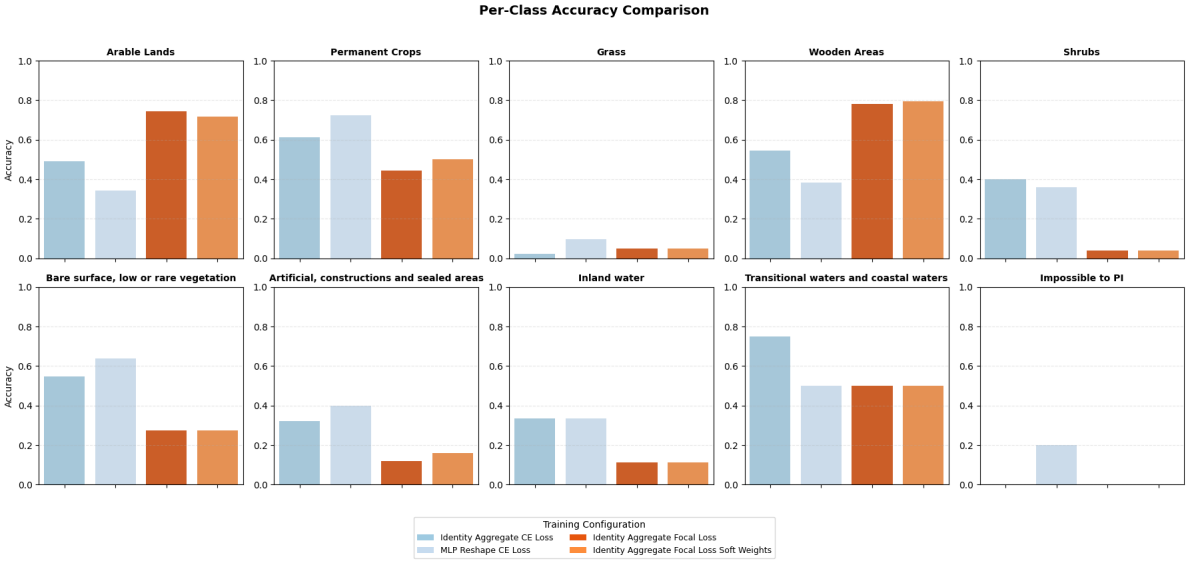


The most distinct behavior observed was the divergence between the Class-Weighted Cross-Entropy (CE) and Class-Weighted Focal Loss objectives.

- **Convergence vs. Balance:** Runs utilizing Focal Loss (Runs 3 & 4) achieved significantly lower test loss values (≈ 0.49) compared to CE (≈ 1.69), indicating stronger optimization of the numerical objective. This translated into higher *Micro Accuracy* ($> 53\%$), as the model effectively maximized performance on dominant classes (e.g., *Wooden Areas*, reaching 78% accuracy).

- **Minority Class Collapse:** However, this optimization came at the cost of the under-represented classes. While Weighted CE maintained a consistent recognition capability across diverse categories (yielding a higher *Macro Accuracy* of 40.2%), Focal Loss tended to ignore difficult minority classes to focus on high-frequency examples. For

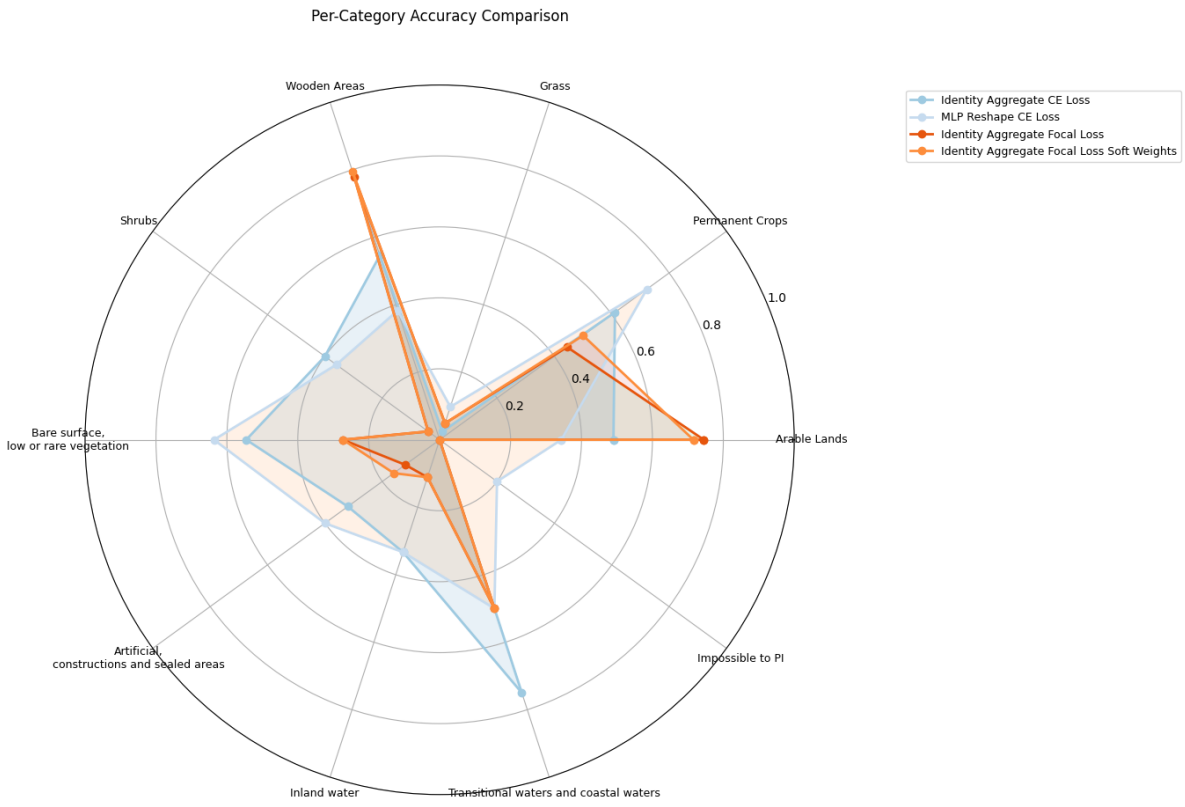
instance, accuracy on *Shrubs* dropped from 40% (CE) to 4% (Focal), and *Artificial Surfaces* dropped from 32% to 12%.



1.2 Decoder Architecture Choice

Contrary to the impact of the loss function, the choice of decoder architecture had a negligible impact on overall performance. The high-capacity **MLP Decoder** (Run 2) did not offer a significant advantage over the lightweight **Identity Decoder** (Run 1), with Macro Accuracy differing by less than 0.5%. This suggests that the pre-trained TerraMind backbone already encodes sufficiently robust spectral features, and a simple linear aggregation is adequate for this classification task.

Other decoders used in test experiments yielded much worse results and are therefore not discussed here.


Per-Category Accuracy Comparison
Wooden Areas
Grass
Shrubs
Permanent Crops
Bare surface,
low or rare vegetation
Arable Lands
Artificial,
constructions and sealed areas
Impossible to PI
Inland water
Transitional waters and coastal waters
0.2
0.4
0.6
0.8
1.0
Identity Aggregate CE Loss
MLP Reshape CE Loss
Identity Aggregate Focal Loss
Identity Aggregate Focal Loss Soft Weights

Conclusion

Our experiments highlighted a critical trade-off between optimization efficiency and class equity when fine-tuning foundation models for classification on highly imbalanced geospatial data. While Class-Weighted Focal Loss successfully minimized the global loss and improved recognition of dominant land covers, it ultimately failed to generalize to the rarest classes, leading to a significant drop in macro-average performance compared to Weighted Cross-Entropy.

Regarding dataset balancing, enforcing equal class sizes via undersampling is strongly ill-advised for this dataset; it would reduce the training set to the size of the smallest minority (approx. 20 samples), thereby discarding the vast majority of valuable spectral variance present in common classes like Arable Lands. A more promising approach for future work would be to investigate synthetic data generation or aggressive augmentation strategies that specifically target underrepresented classes. This could also be achieved by downloading much larger datasets to make sure that the model is exposed to at least 100 images per class, although this would require significantly higher computational resources. Extending the hyperparameter search to tune the focusing parameter (γ) of the Focal Loss could potentially identify a stable configuration that balances convergence speed with the retention of minority class accuracy.

Bibliography

[1] European Commission, Eurostat. Lucas: Land use and coverage area frame survey, 2024. Accessed: 2025-12-06.

[2] Johannes Jakubik, Felix Yang, Benedikt Blumenstiel, Erik Scheurer, Rocco Sedona, Stefano Maurogiovanni, Jente Bosmans, Nikolaos Dionelis, Valerio Marsocci, Michael Kopp, et al. Terramind: Large-scale generative multimodality for earth observation. *arXiv preprint arXiv:2504.11171*, 2025.